Efficient Alignment of Mini-LLMs for Hausa Language

Using GRPO and Unsloth for Digital Sovereignty

Faisal Muhammad Adam

ACETEL - National Open University of Nigeria

2026

The Problem: Linguistic Shadowing

- **Density Gap:** High-resource languages can dominate generation in low-resource contexts.
- **Cross-lingual Drift:** Base models drift to non-target syntax when Hausa signal is weak.
- **Hardware Barrier:** Traditional RLHF pipelines are expensive for regional labs.

Theoretical Solution: Why GRPO?

Group Relative Policy Optimization (GRPO)

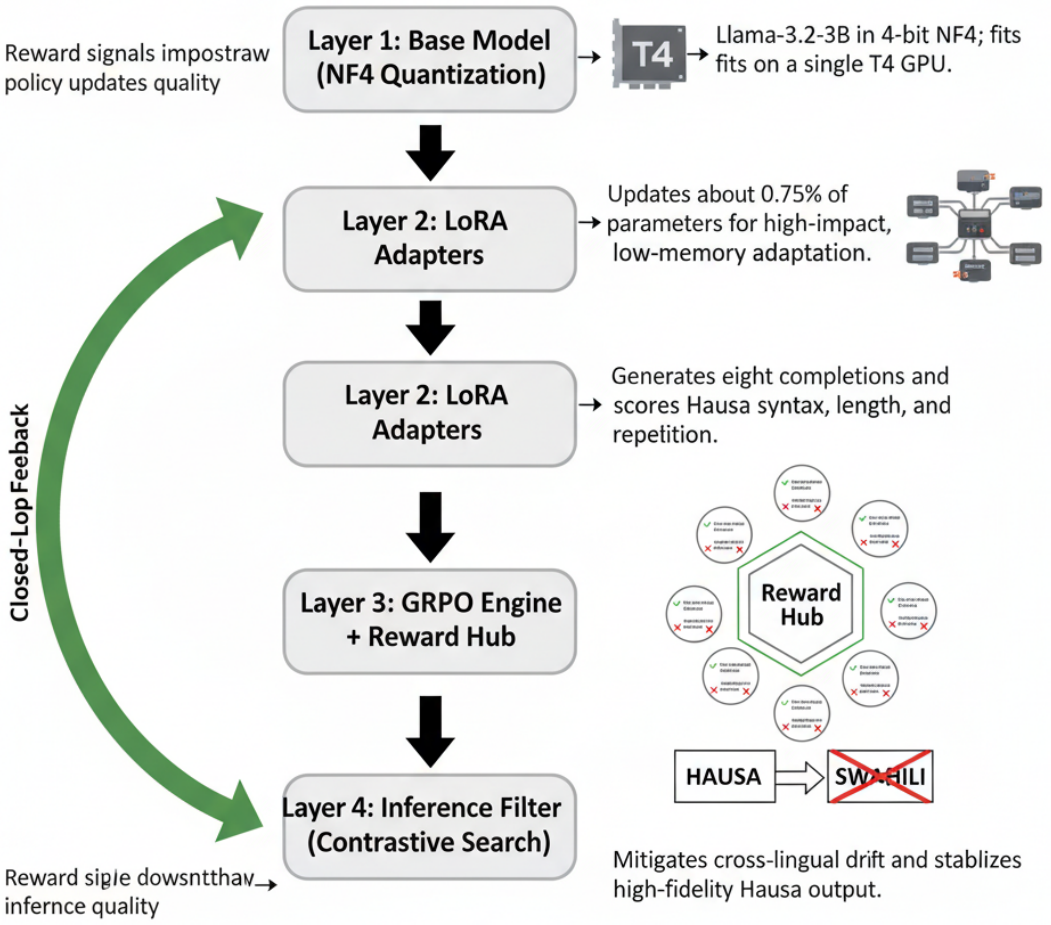
GRPO removes the need for a separate critic model and improves memory efficiency.

$$\hat{A}_i = \frac{r_i - \text{mean}(r_1, \ldots, r_G)}{\text{std}(r_1, \ldots, r_G)}$$

- Relative advantage is computed over a group of $G = 8$ completions.
- Enables practical alignment on a single NVIDIA T4 setup.

System Architecture

Figure 1: System architecture of the proposed GRPO-based Haugn-ment framework



Results: From Collapse to Alignment

Metric	Base	Aligned	Gain
Hausa Syntactic Score	0.42	0.93	+121%
Response Coherence	62%	88%	+26%
VRAM Consumption	14.8 GB	8.2 GB	-44%

Table: Performance benchmarks on Llama-3.2-3B

Conclusion and Impact

- **Democratization:** High-fidelity alignment on limited hardware is achievable.
- **Sovereignty:** Practical roadmap for localized Hausa AI systems.
- **Method:** Mini-LLMs + GRPO + contrastive decoding is a robust low-resource strategy.

Key References

- DeepSeek-AI (2024). *DeepSeek-V3 Technical Report*.
- Hu et al. (2022). *LoRA: Low-Rank Adaptation of Large Language Models*.
- Dettmers et al. (2023). *QLoRA: Efficient Finetuning of Quantized LLMs*.
- Meta AI (2024). *The Llama 3 Herd of Models*.
- Unsloth AI (2024). *Unsloth: Lightweight and Fast LLM Training*.
- Su et al. (2022). *A Contrastive Framework for Neural Text Generation*.

Thank you.